# ODEs

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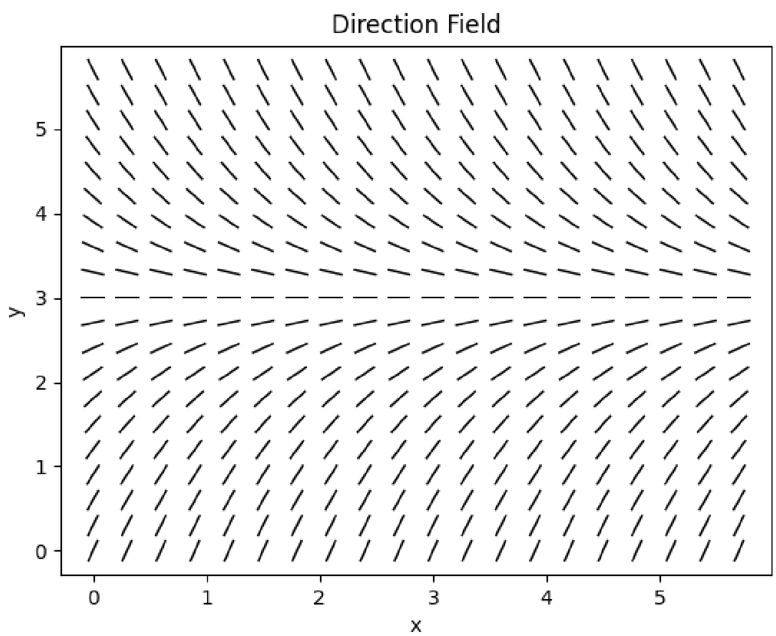
## 1 Section 1.1

### Problem 1

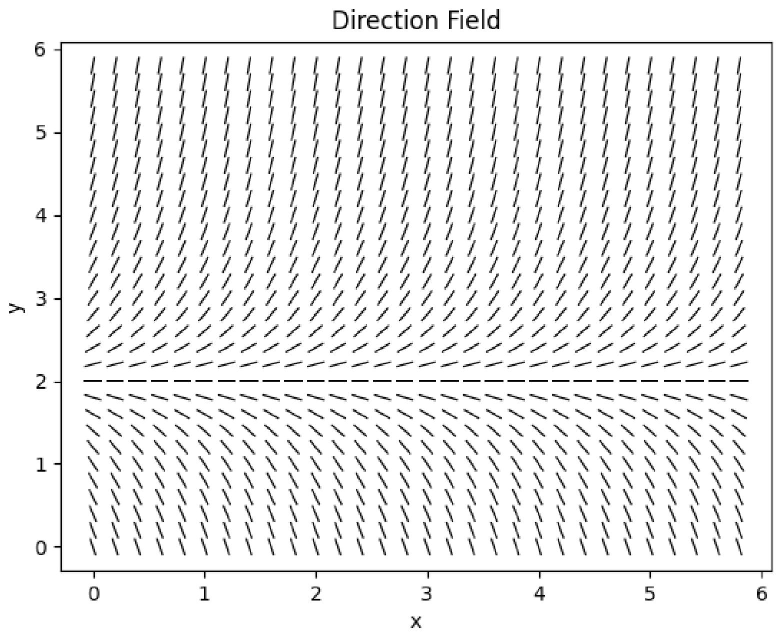
Below shows the direction field for $y' = 3 - y$.




It can be seen that for $y > 3$ the slopes are negative so solutions decrease and for $y < 3$ the slopes are positive so solutions increase. The equilibrium position appears to be at $y(t) = 3$, to which all other solutions converge.

### Problem 2

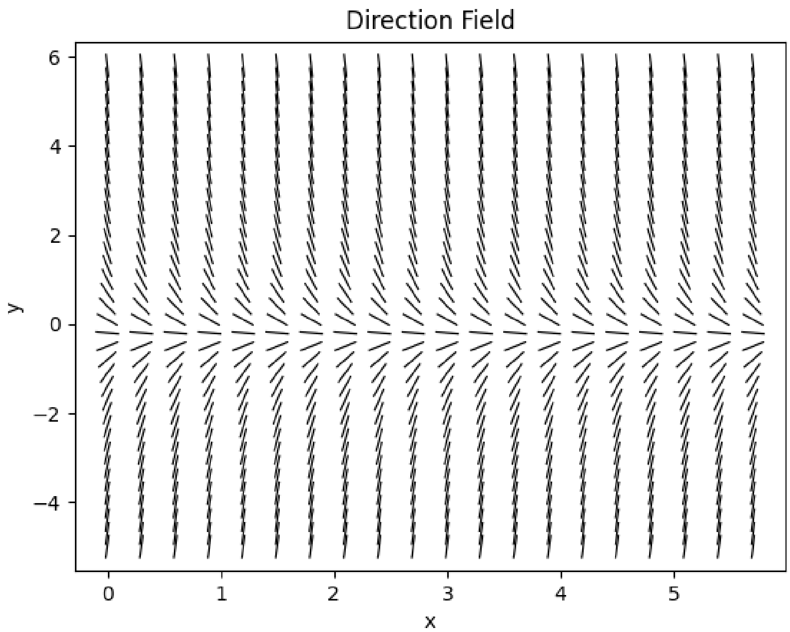
Below shows the direction field for $y' = 2y - 4$




For $y > 2$ the slopes are positive so solutions increase and for $y < 2$ the slopes are negative so solutions decrease. The equilibrium solution appears to be $y = 2$ and all other solutions diverge away from this solution.

## Problem 3

Below shows the direction field for $y' = -1 - 4y$


Direction Field
6
4
2
0
−2
−4
y
0
1
2
3
4
5
x